

I.A.B

Ivan Bagaturiya

Architect

Full portfolio

This is the full portfolio of Ivan Bagaturiya.





02

Profile

Architect ETH Zurich. Specializing in computational design and digital workflows. His work bridges architecture and software through parametric modeling, scripting, and process automation. He develops tools and systems that make complex design processes more efficient, scalable, and intelligent.

01

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- github.com/ibagaturiya
- instagram.com/ibagaturiya
- Zürich, CH

07

Languages

- Deutsch
- Schweizerdeutsch
- Englisch
- Russisch
- Französisch

03

Education

Master of Science ETH Architektur

2023 bis 2025

ETH Zürich

PROFESSORS Prof. Dr. Pauli Prof. Charbonnet
Prof. Heiz Prof. Gramazio
Prof. Kohler Prof. Emerson

Bachelor of Science ETH Architektur

2019 bis 2023

ETH Zürich

PROFESSORS Prof. Kerez Prof. Rossi
Prof. Gagliardi Prof. Hovestadt
Prof. Christ Prof. Gantenbein
Prof. Spiro Prof. Deplazes

04

Experience

Kurator und Dozent Digitales Entwerfen

2025 bis heute

ETH Studio Foundations und Gramazio Kohler Research

- Koordination der interdisziplinären Zusammenarbeit im ersten Jahr des Bachelorstudiums Architektur mit acht Professuren und mehr als 350 Studierenden.
- Entwicklung von Tutorialserien und Plugins für Rhino, die von Studierenden aktiv genutzt werden.
- Durchführung von Übungen und Vorlesungen zu Rhino, Grasshopper und weiteren digitalen Werkzeugen.
- Verantwortung für Infrastruktur, Materialverwaltung und Prozessautomatisierung des Studios.

Consulting Digitale Planung, Spezialisierung Vectorworks

2025 bis heute

HMQ AG – Chur

- Optimierung von Arbeitsprozessen mit BIM sowie beim Datenaustausch zwischen unterschiedlichen BIM Anwendungen.
- Qualitätssicherung von Projektmodellen und Dokumentation vor der Abgabe an Kundinnen und Kunden.

Design und Fabrikationsassistent

2024 bis 2025

ETH Startup impact.build

- Entwicklung und Optimierung von Prototypen für Konstruktionen aus Lehm und Holz mit Robotern für innovative und nachhaltige Bauweisen.
- Erstellung parametrischer Modelle für Entwurf und Fertigung.
- Verantwortung für die Ausführung digitaler und handwerklicher Bauprozesse.

Computational Design Hilfsassistent

2023 bis 2024

ETH Digital Building Technologies, Professur Dillenburger

- Betreuung von rund 15 Studierenden bei Arbeitsprozessen mit Grasshopper und Rhino.
- Auswertung und Besprechung von Resultaten.

Architekt, Visualisierung und IT Support

2021 bis 2024

Studio Hajnos GmbH

- Optimierung bestehender Entwurfsprozesse in 3D und Einführung von BIM zur Steigerung von Effizienz und Kundenzufriedenheit.
- Erstellung von Projektplänen und hochwertigen Visualisierungen.
- Entwicklung und Umsetzung von Anwendungen für VR.
- Interner Ansprechpartner für Arbeitsprozesse in 3D und VR sowie technische Unterstützung von Mitarbeitenden und Praktikanten.
- Betreuung und Einführung neuer Praktikanten.

Skills & software

Software

Rhino Grasshopper Python RhinoCommon Visual Studio Dynamo AI for BIM
 Vectorworks Archicad Revit Navisworks Solibri BIM 360 Twinmotion D5 Blender
 Radii Adobe Creative Suite Affinity Suite

Architecture and BIM

Architectural design Computational design Parametric design BIM Digital workflows
 Model quality assurance Interdisciplinary coordination Design and fabrication

Programming and automation

Python programming Visual programming Tool development Plugin development
 Process automation Data management Workflow analysis Debugging

Visualization, XR and capture

Visualization Virtual Reality VR XR 360 degree photography Drone photography LiDAR
 Gaussian Splatting

Teaching and operations

Teaching Tutorial development Lecture delivery Studio coordination
 Infrastructure management Material management Technical support Mentoring

Interests & achievements

Fotografie

- 360 Grad Fotografie
- VR, AR und XR
- Drohnenfotografie und -filmografie
- Eventfotograf der Tribute-Band von 'Ramones', 2019 bis 2022
- Analoge und digitale Fotografie, professionelle Porträts, Objektfotografie und Makrofotografie
- Preisträger des 'National Geographic' Fotowettbewerbs 2015

Musik

- Jurymitglied bei Jugend musiziert, 2018 bis 2021
- Mehrfacher Sieger bei Jugend musiziert auf Bundesebene: E-Bass Solo, E-Gitarre, Akustikgitarre und Schlagzeug
- Tontechnikassistent bei musikalischen Events in Prag, Stockholm, Helsinki, Paris und Moskau, 2014 bis 2021

Kommunikation

- Silberpreis beim China International College Students' 'Internet+' Innovation and Entrepreneurship Wettbewerb 2021
- Rhetorik- und Redekurse 2015

Zeichnung und Grafik

- Ausstellung in der Smolenskaya Galerie, Moskau 2017: Öl, Aquarell, Bleistift und Modellbau
- Ausstellung in der Kunsthalle Zürich: Mycelium As Building System, Between Design Iterations And Living Mockups

Sport

- Schwimmen
- Eishockey, Captain 2016 bis 2019
- Vietnamesische Kampfkunst Tam Qui
- Volleyball, Europaspiele 2016 in Brüssel

Selected projects

_machbarkeitsstudie	05
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0062 / 01

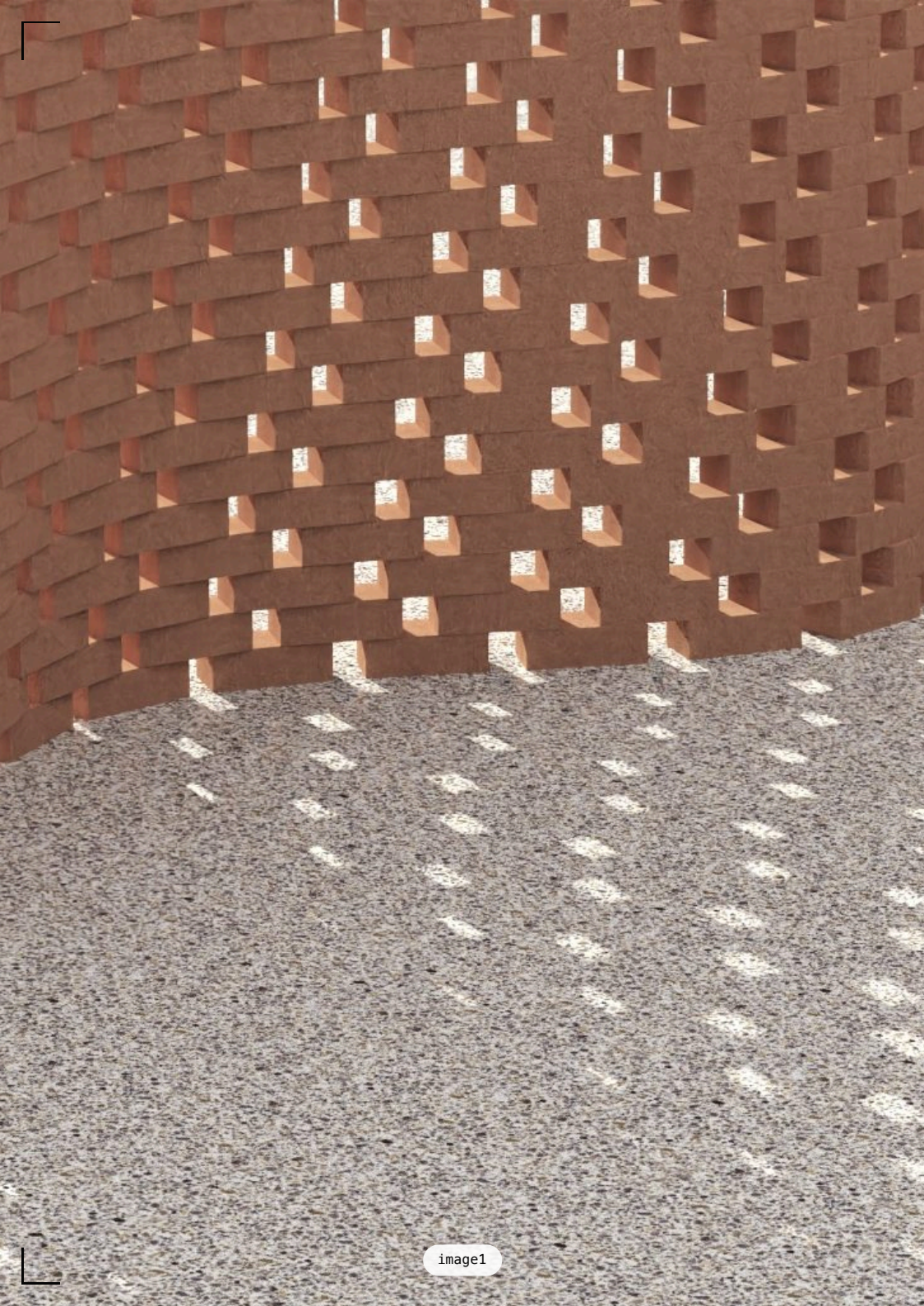
_machbarkeitsstudie

2026

Feasibility study – an interactive tool for
maximum legal building volume in Kanton
Zürich.

SKILLS USED

#tech



0056 / 02

_sinuswall

2025

This project shows how a parametric tool can keep a sinusoidal brick wall buildable before the brick size is known.

SKILLS USED

#shapediver

#computational-design

#digital-workflow

image1



0055 / 03

_der Leuchtturm

2025

This project shows how ten slats, a rope, and a sheet of textile can become a lighthouse in 45 minutes.

SKILLS USED

#architectural-design

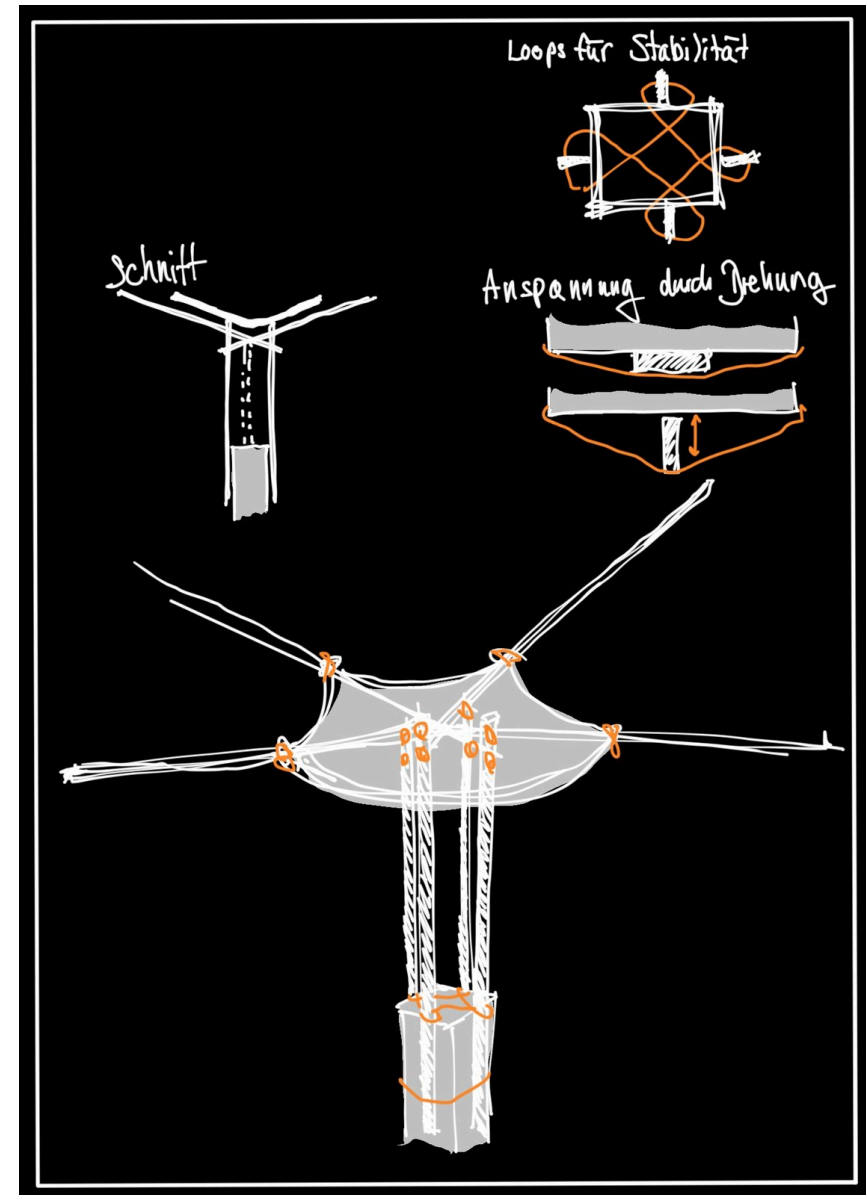
#prototyping

#teaching

0010 image

we were asked to test the first task for the 1st year Architecture BSc Students, that they will have to do in a Day. We got only one hour. The task was to build a shelter. we got only textile, beams and rope.

0011 abstract



0020 image



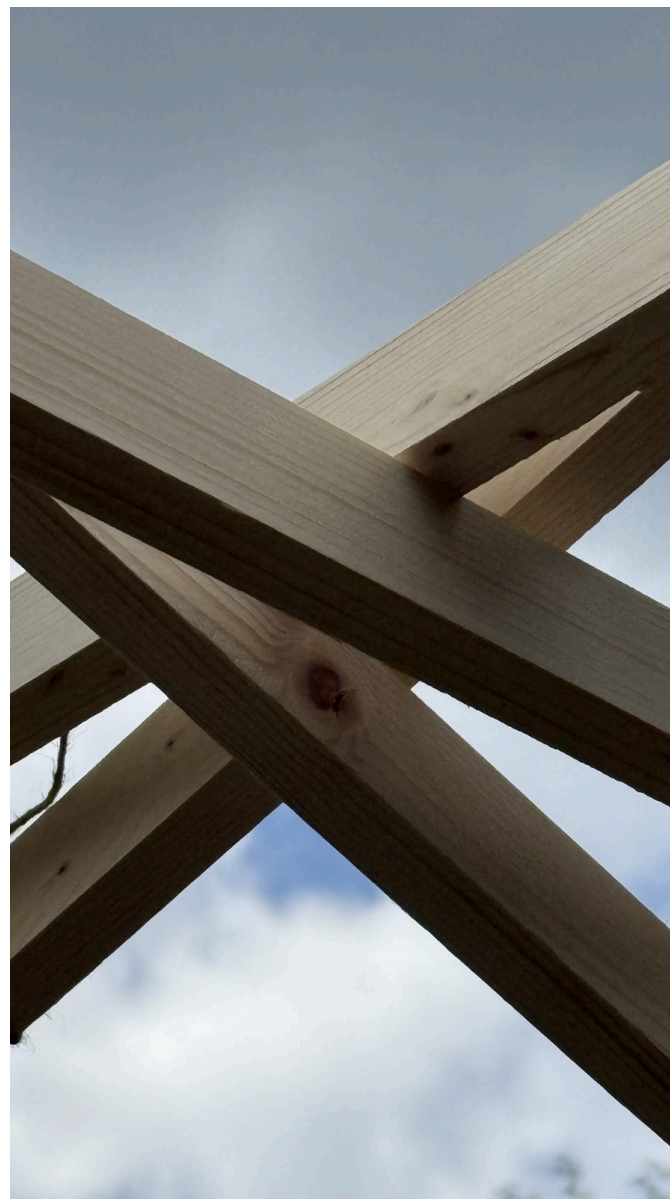
0030 image

and here some details

0035 abstract



0040 image



0050 image



0054 / 04

_the NOTE

2025

This project shows how computational design can turn an underground parking garage into a concert hall.



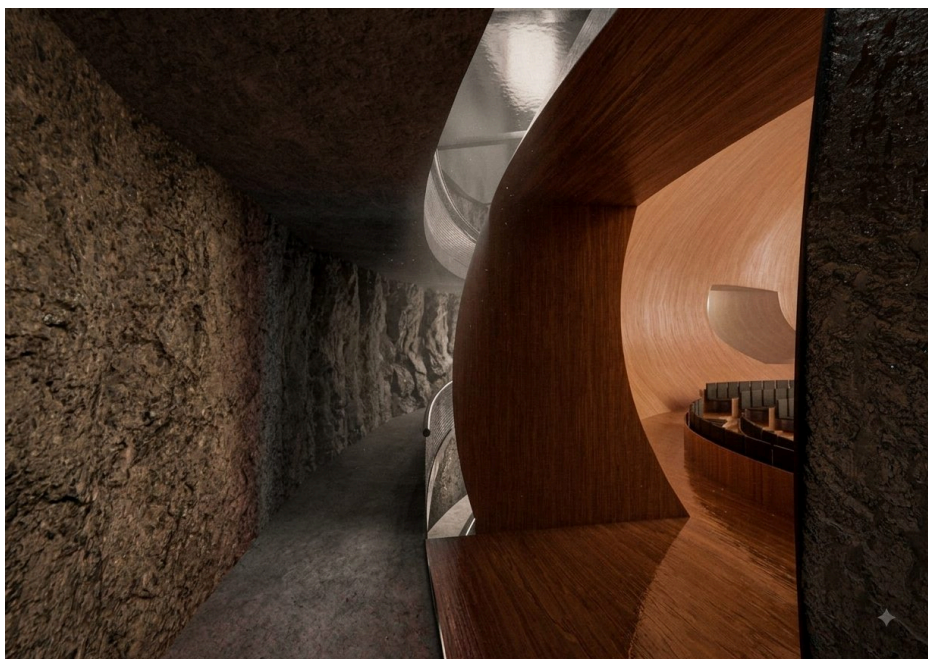
SKILLS USED

#grasshopper

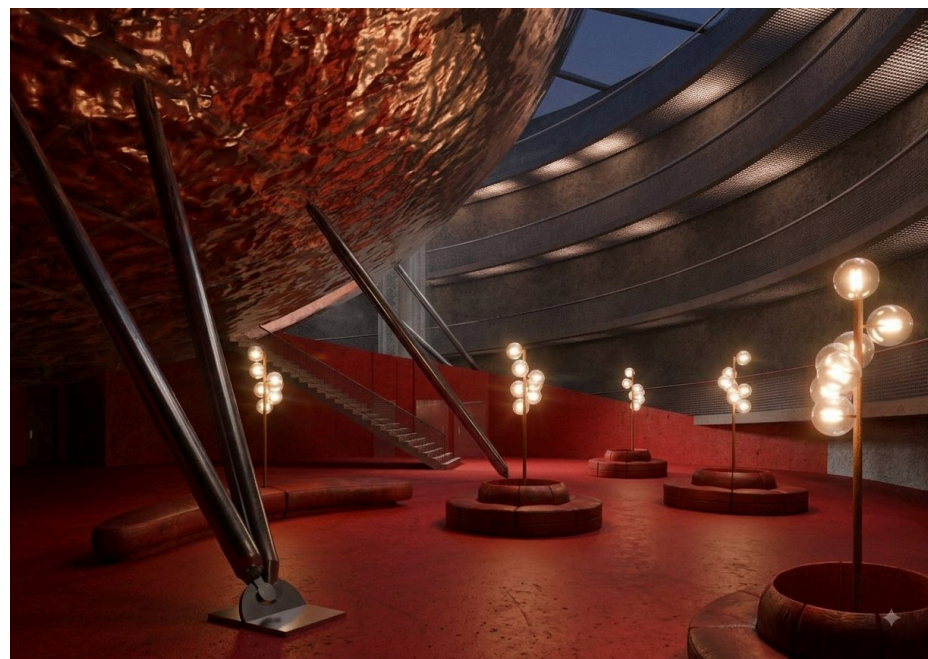
#architectural-design

#parametric-design

#spatial-organization



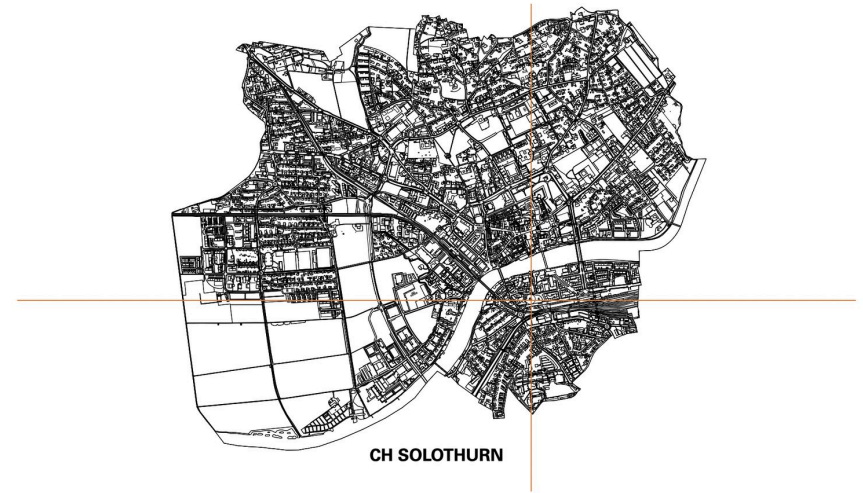
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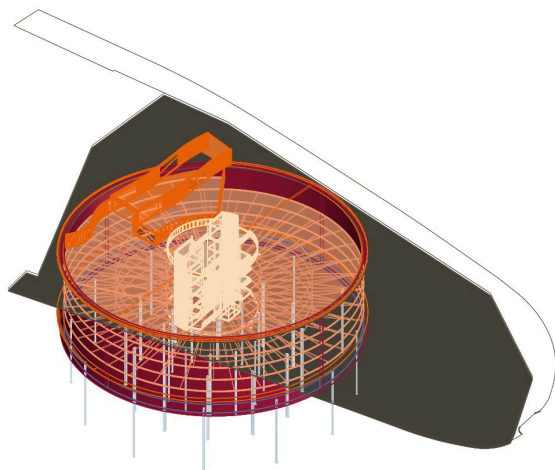
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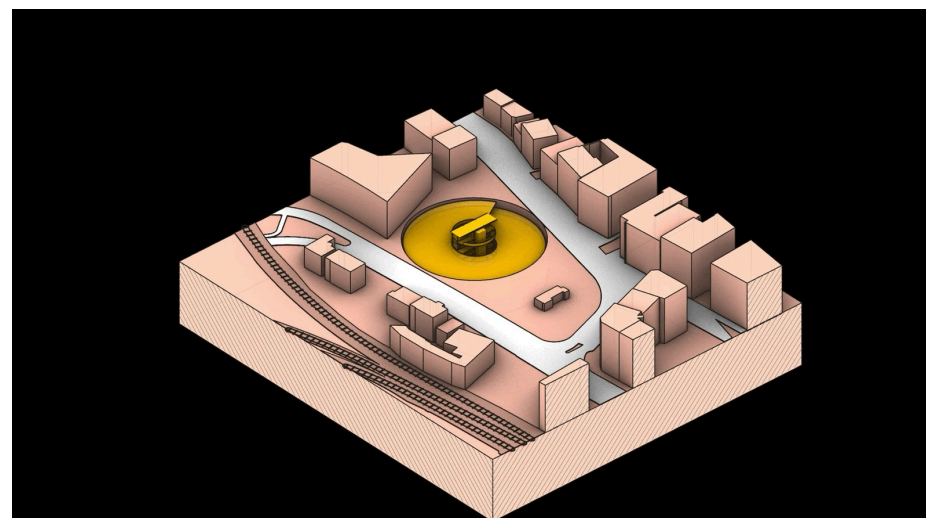
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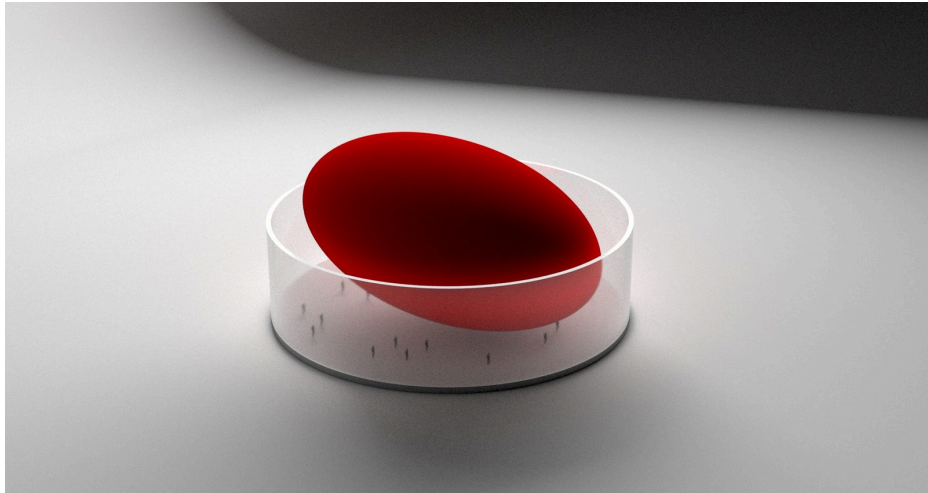
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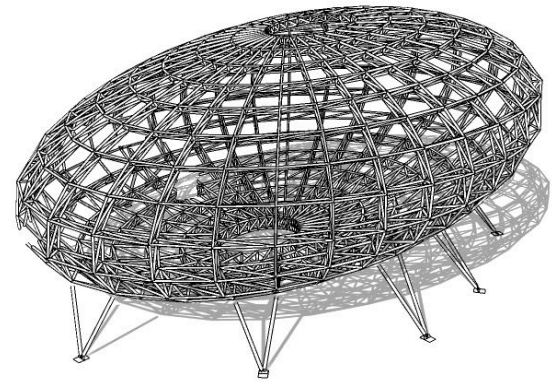
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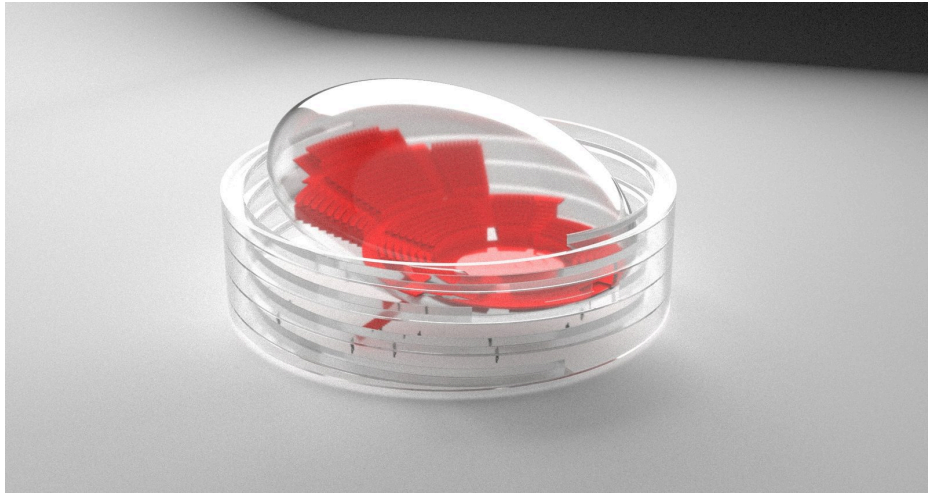
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00016image



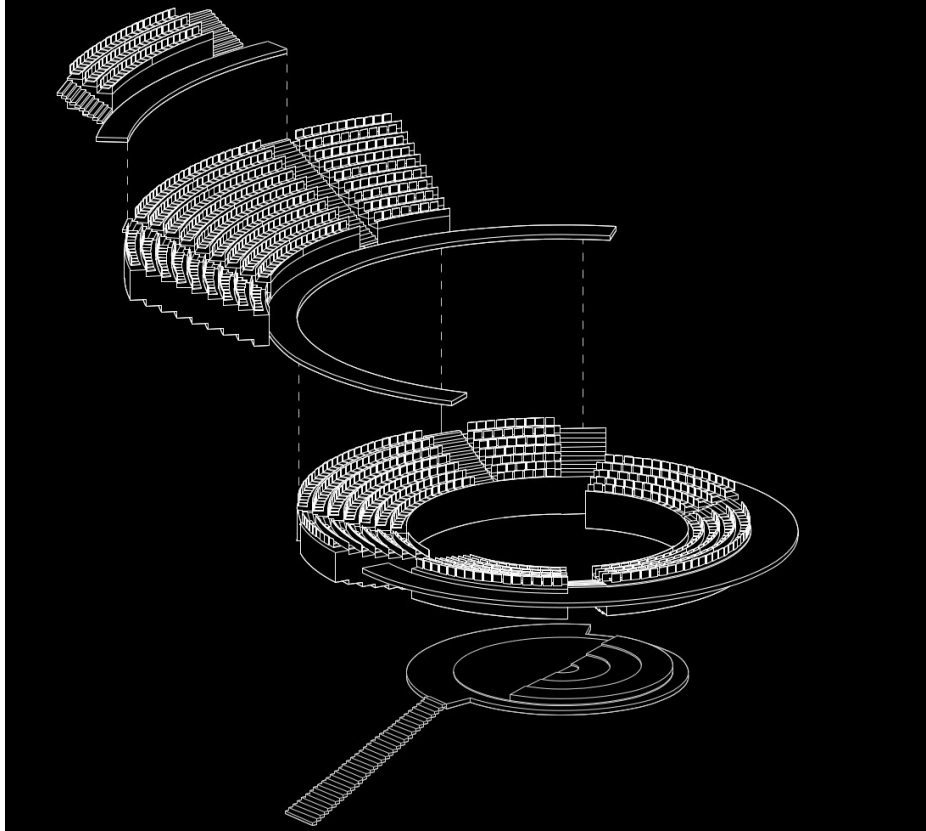
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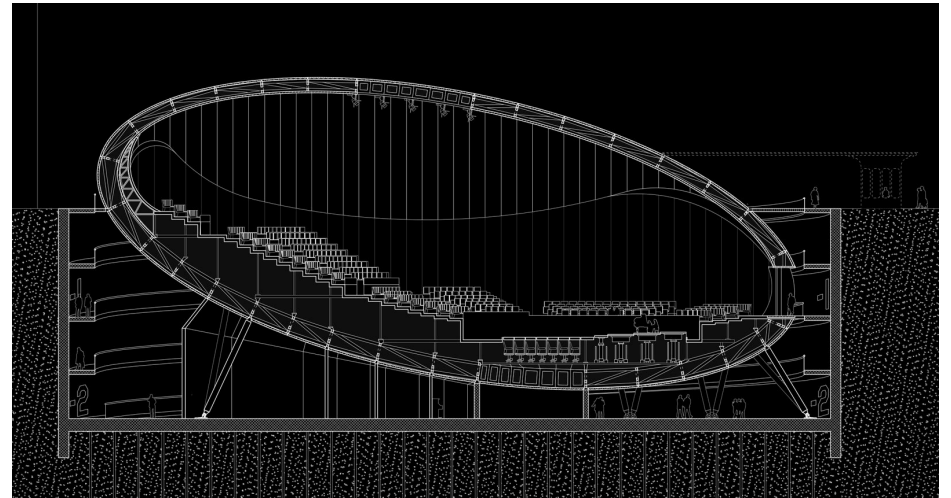
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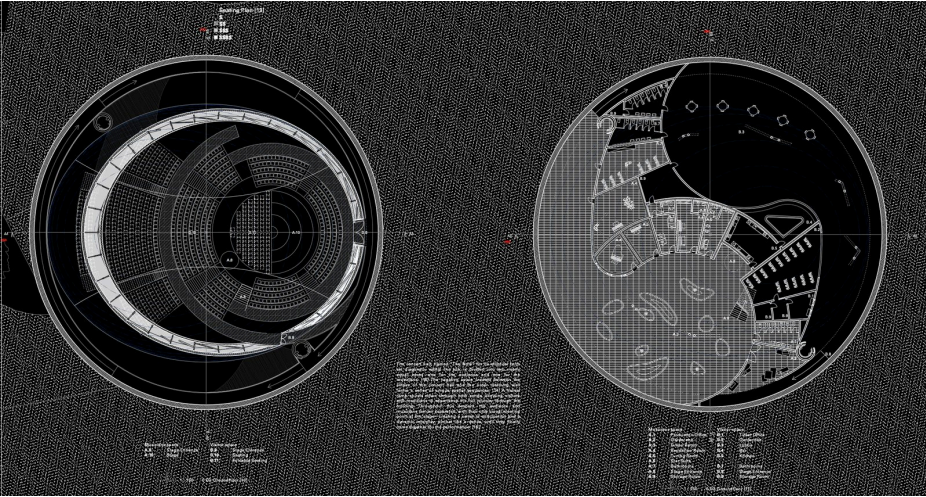
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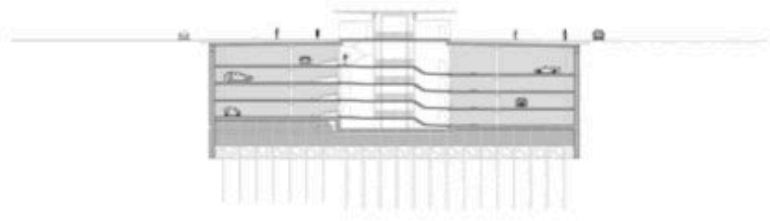
00020image



00021image



00022image

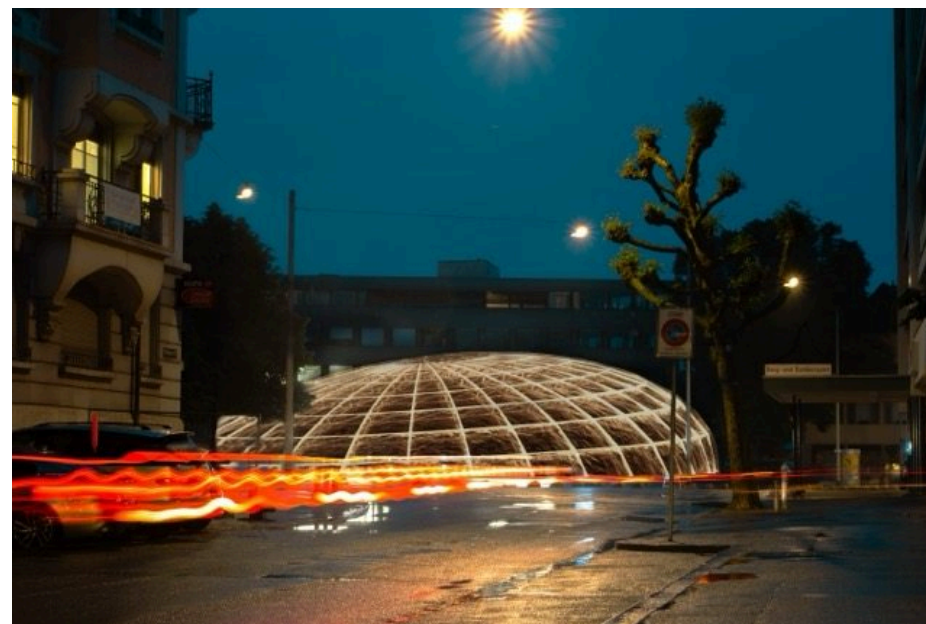


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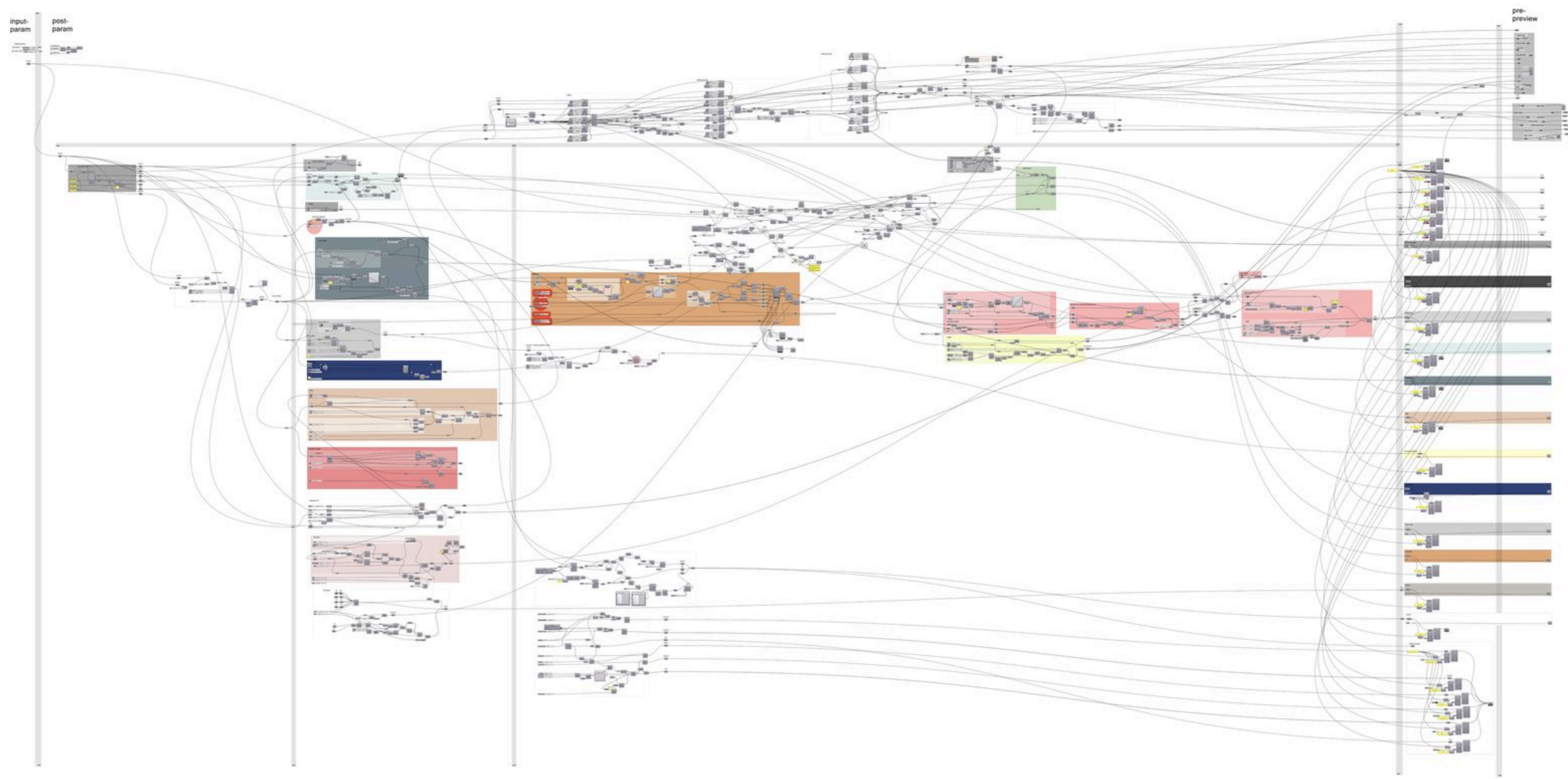


10 FPS

00024imagesec



00025image



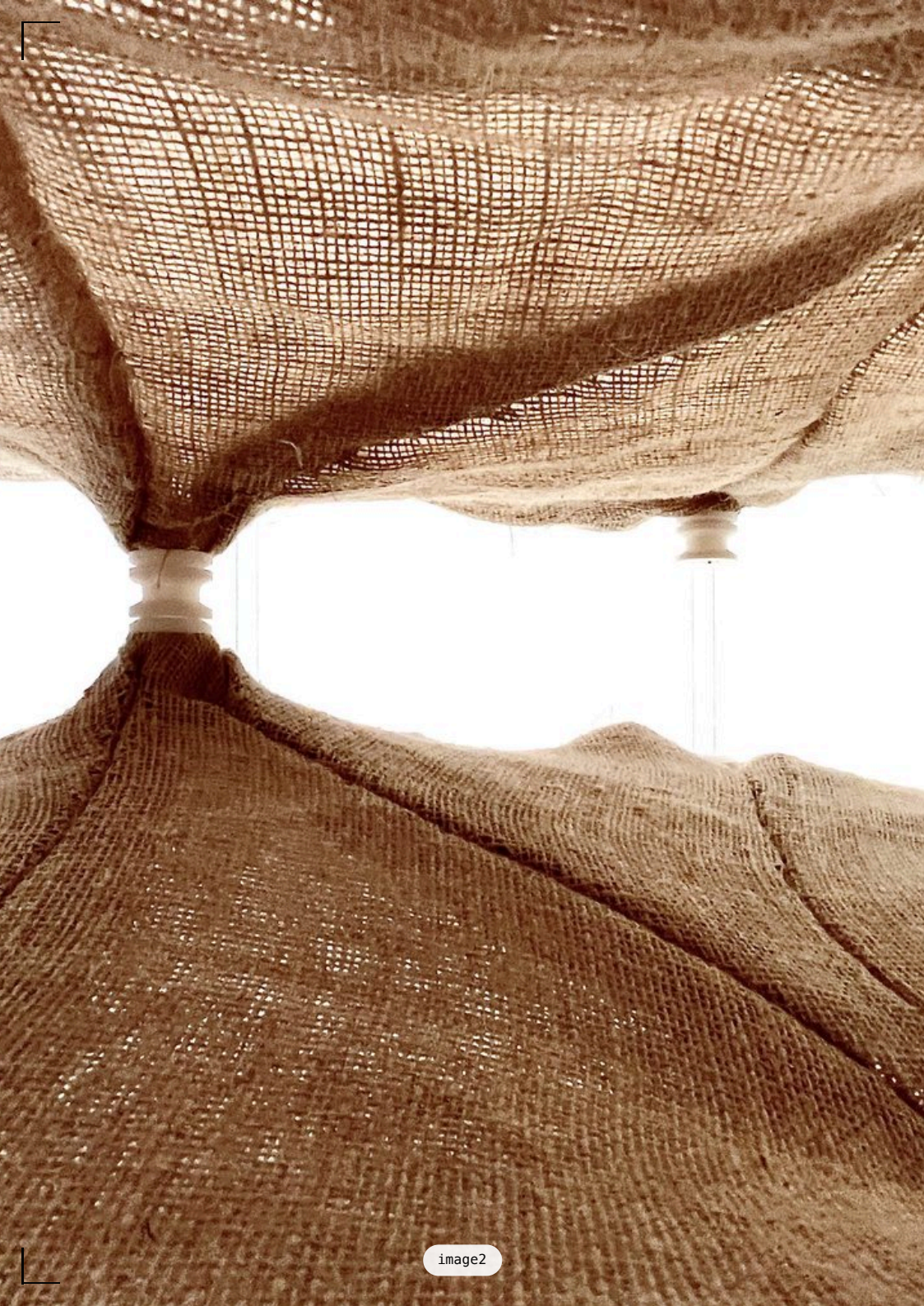


image2

0052 / 05

_Zwischendecken

2024

This project shows how mycelium and parametric form-finding can turn soft textile into a structural vault.

SKILLS USED

#grasshopper

#parametric-design

#material-research

#prototyping



image3

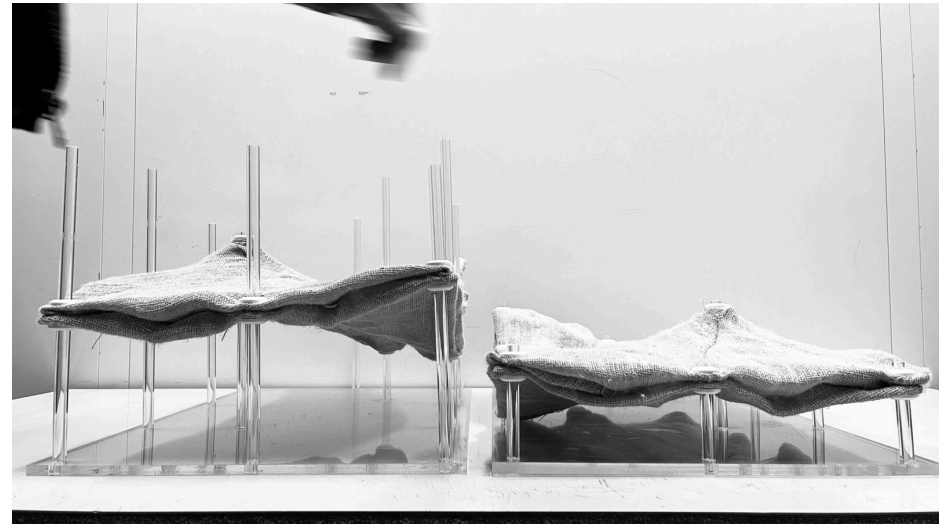


image4



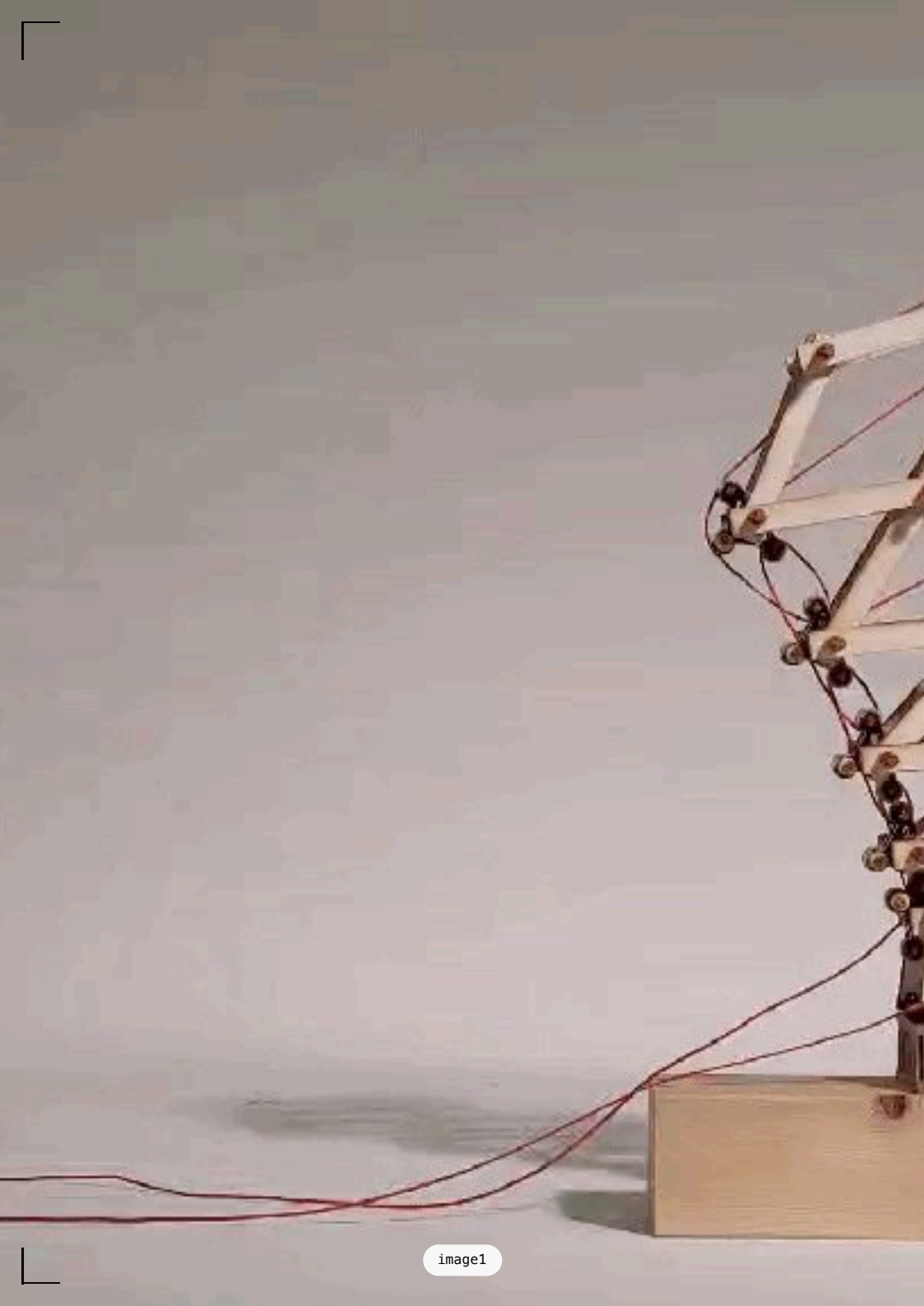


image1

0051 / 06

_the CABRIO

2024

This project shows how parametric architecture can turn a noisy parking lot into an adaptable urban stage.

SKILLS USED

#rhino

#grasshopper

#parametric-design

#urban-design

#acoustic-design





image3



image4



image5

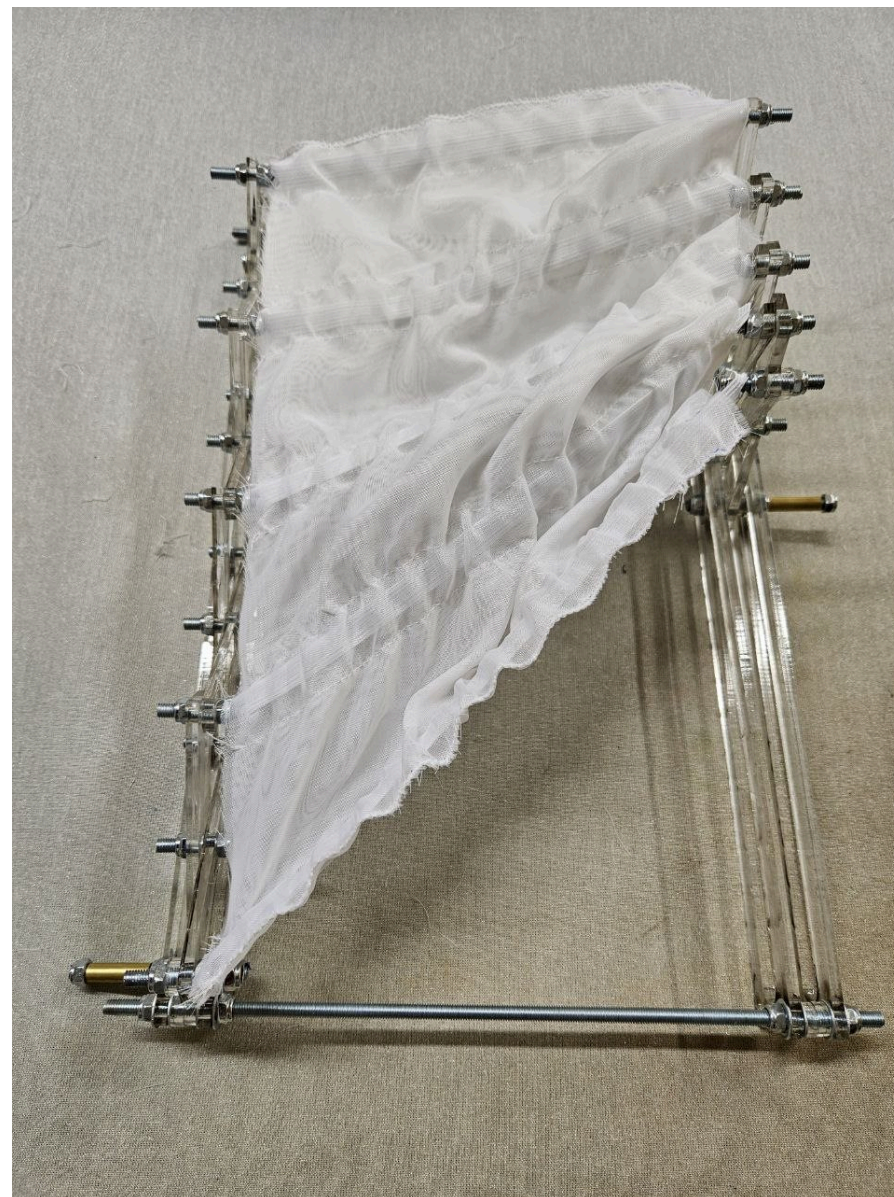


image6





0050 / 07

_panoramortraits

2024

This project shows how 36 analog exposures can turn a changing room into a continuous archive of space, time, and memory.

SKILLS USED

#photography

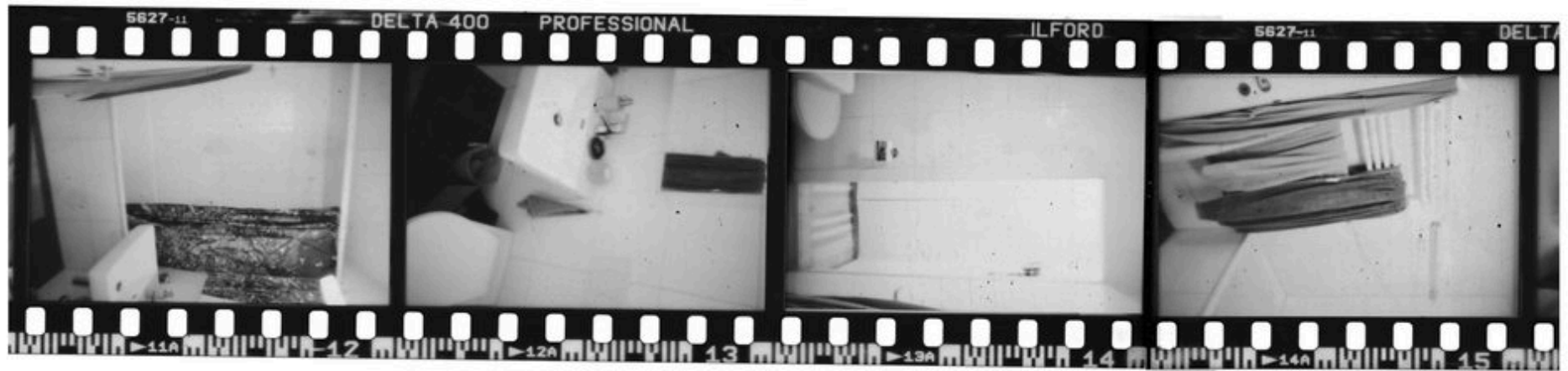
#visual-storytelling



image2



image3



0015 / 08

_Infrastructure as storage

2022

what if we would use the infrastructure as storage space? and what if this infrastructure would be our facades?

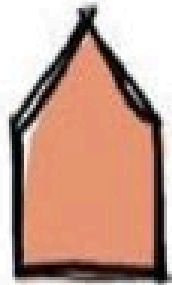
SKILLS USED

#architectural-design

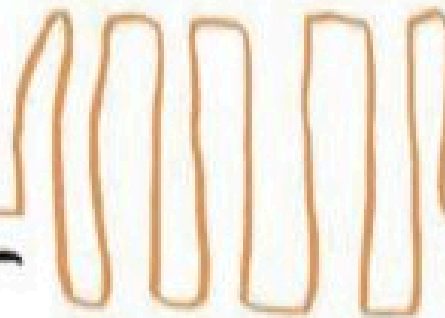
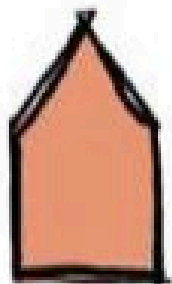
#infrastructure

#adaptive-reuse

RESTAURANT

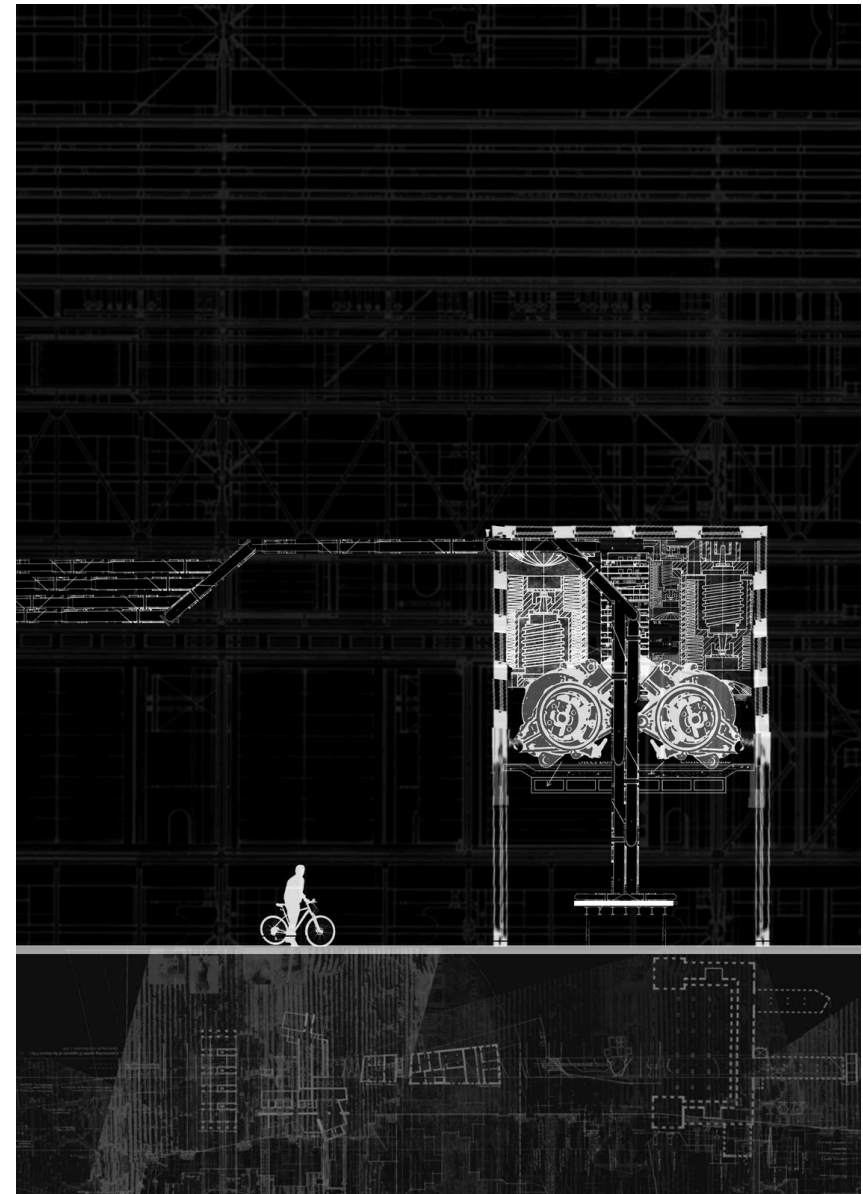


HAUS

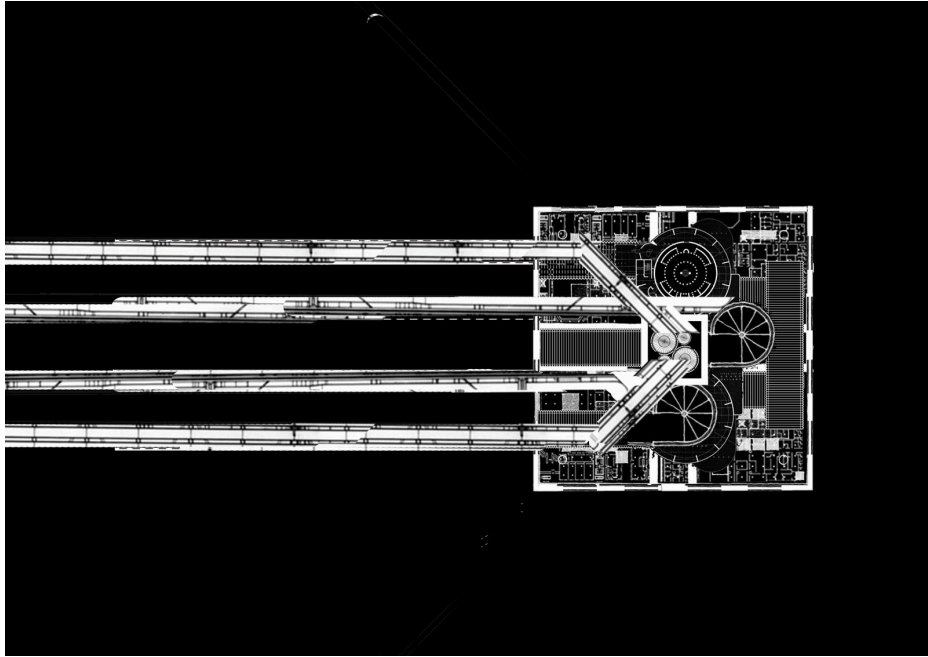


This just-in-time principle leads to congested city streets by cars, bike couriers etc. A new infrastructure is introduced that reduces traffic on the roads and takes over the storage and delivery of small goods. The infrastructure is connected to the rail network and has links to the center of the city as well as the outskirts. The railway stations are connected and provide the link between the otherwise linear strands.

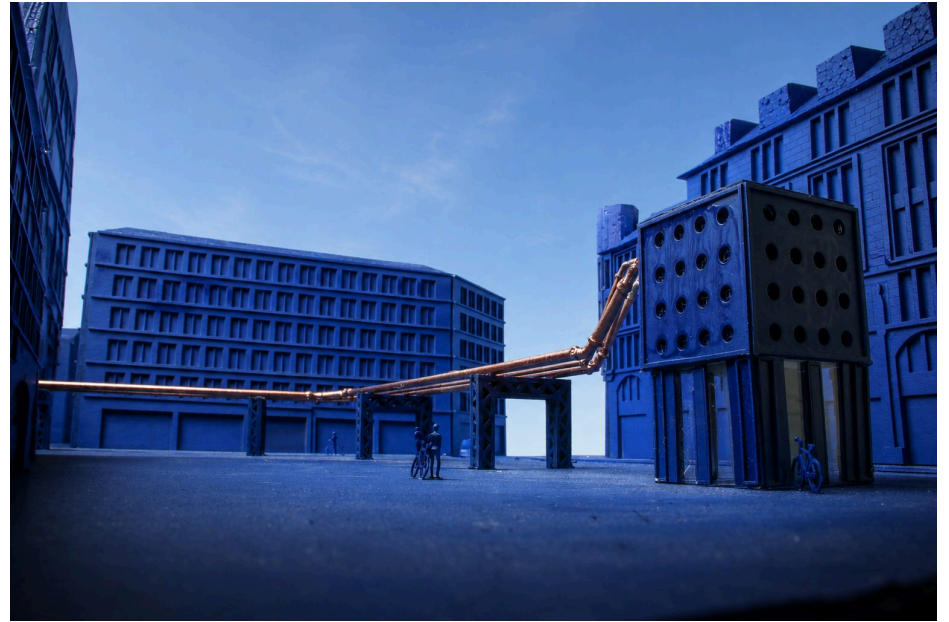
0020_abstract 2



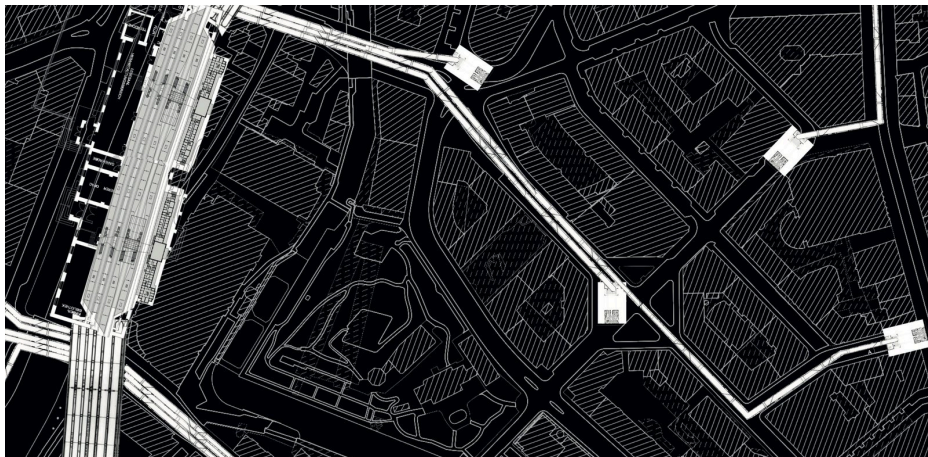
0021_image



0022_image



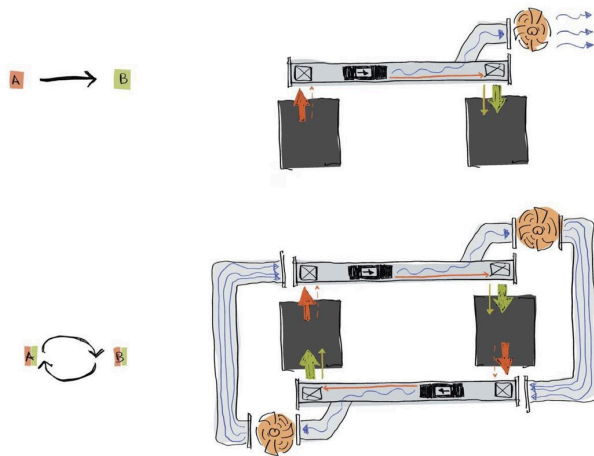
0023_image



0024_image

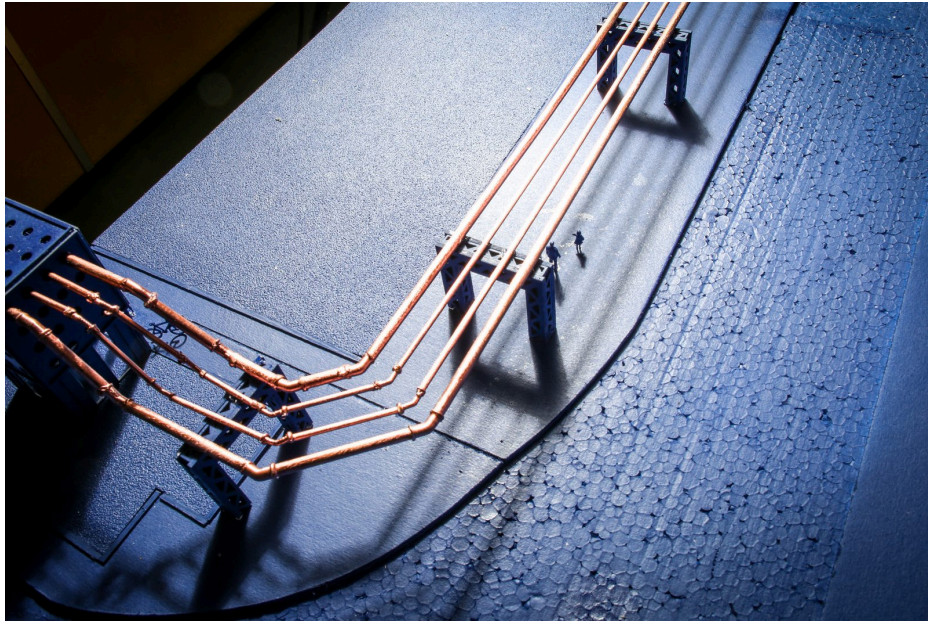
Goods are transported by vacuum through tubes. At the end of the tube there is a checkpoint. This consists on the one hand of a machine room with the technology to drive the infrastructure and on the other hand of a kiosk for receiving and delivering the goods.

0025_abstract 22



Goods will be transported by continuous vacuum through tubes.

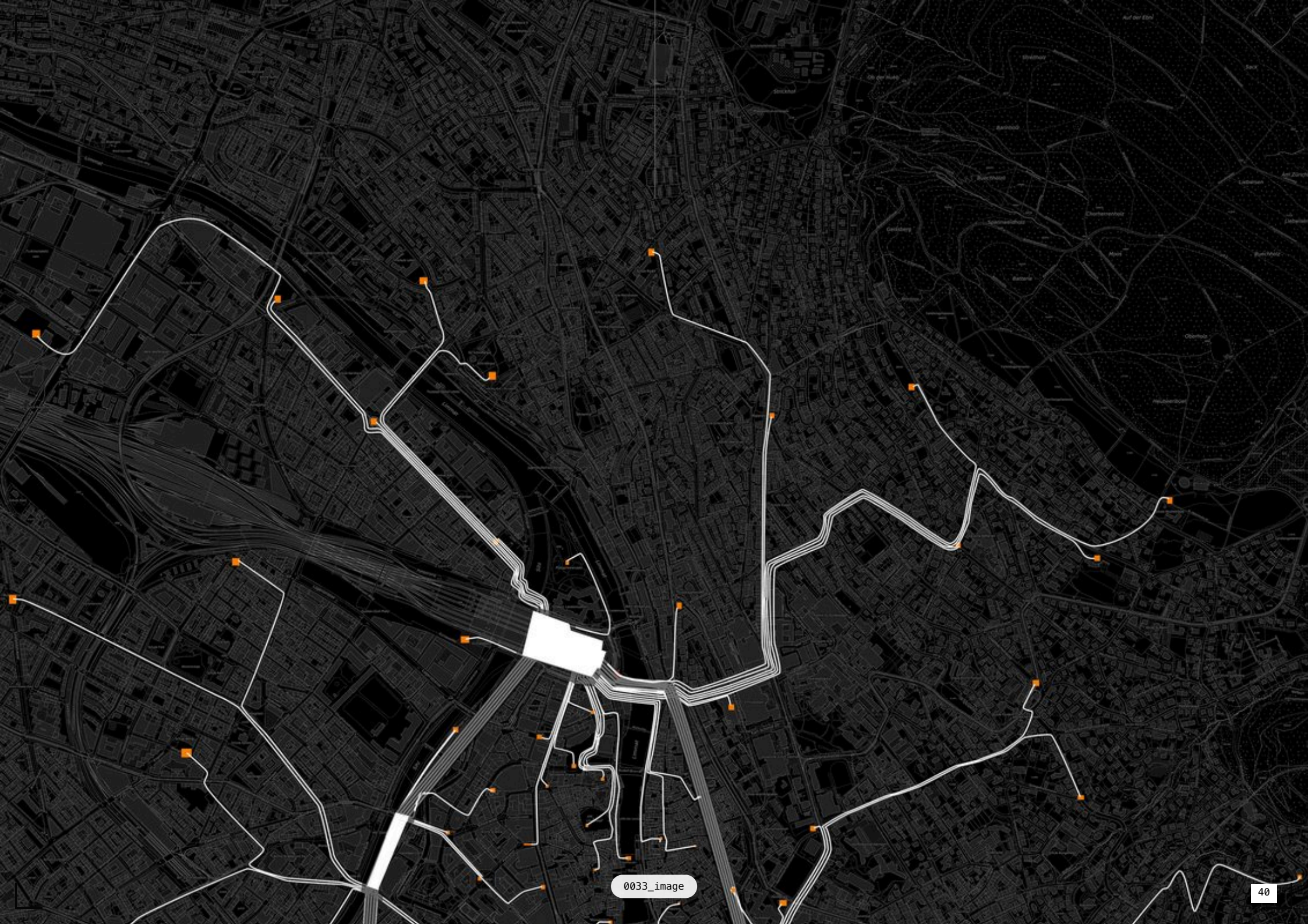
From the checkpoints there is a fine distribution, where the orders are brought to the individual addresses by bicycle. Thus, there are only a certain number of couriers who take care of the deliveries of all the shops in the vicinity of a checkpoint. The infrastructure covers the busy roads, so the goods are transported through the tubes avoiding congestions, instead of adding couriers to the traffic. The system works in both directions, so the infrastructure runs in a cycle. It relieves the strain on traffic and enables goods to be transported quickly from the city center to the outskirts and vice versa.



0030_image



0031_image





0013 / 09

_Wissen Hütten

2020

Project description forthcoming.

SKILLS USED

#architectural-design

#timber-construction

#model-making

Subject Design and Construction III
 Date HS - 3th Semester
 Name Ein neues gta - Archiv in der Hufabrik 4
 Project Participants Zeming Li, Ivan Bagaturiya

Following the concept, the degree of publicity of the uses in the building is designed by the construction and the lighting.

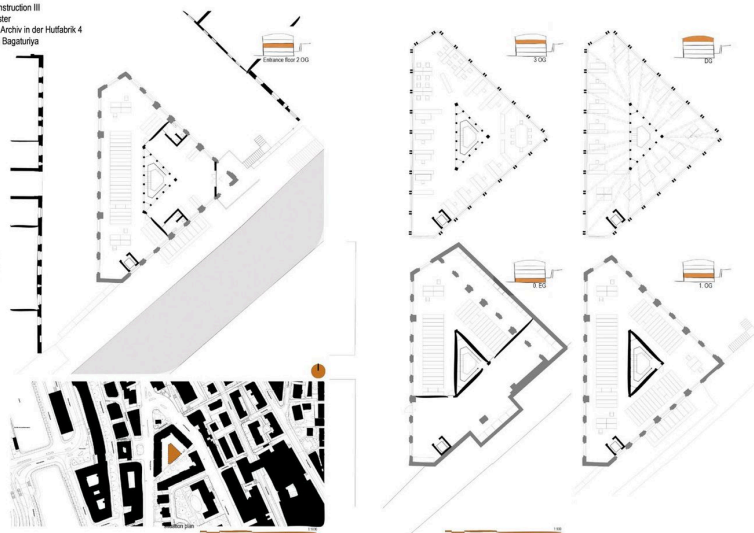


image2

Our concept is to create a light, airy floor in which most of the people's activities will take place. In contrast, the heavy archive materials are stored in the existing solid building.

All floors are connected with an atrium that penetrates the building in the form of a triangle and also serves as THE only stairwell for the building. Thanks to the upper glass covering, the atrium becomes a second natural source of light.



image3



image00002

0010 / 10

_Areal Archive

ongoing

An evolving aerial archive that reveals familiar landscapes from an unfamiliar point of view.

SKILLS USED

#drone-photography

#aerial-photography

#visual-storytelling



image0003

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